

TITANS-SUKKUR

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Backend Development — Lecture 1

Registration API using **Node.js** + **Express** + **Postman**

Learning Objectives

Lecture complete hone ke baad student:

- ▶ Node.js project create kar sake.
- ▶ Express install kar sake.
- ▶ Server create kar sake.
- ▶ Middleware samajh sake.
- ▶ POST API bana sake.
- ▶ GET API bana sake.
- ▶ Postman se API test kar sake.

Step 1 Create Folder

```
mkdir Backend  
cd Backend
```

Step 2 Initialize Project

```
npm init -y
```

Ye command package.json create karti hai.

Step 3 Install Express

```
npm install express
```

Step 4 Create server.js

```
// Express package ko import kar rahe hain  
const express = require("express");
```

```
// Express application create kar rahe hain
```

```

const app = express();
// Server kis port par chalega
const PORT = 3000;

// Middleware
// Ye incoming JSON data ko read karne ke liye use hota hai
app.use(express.json());

// Temporary array
// Isme users ka data temporarily store hoga
// Server restart hoga to sara data delete ho jayega
const users = [];

// =====
// Registration API
// =====

// POST request register endpoint par receive hogi
app.post("/register", (req, res) => {

  // Request body se name, email aur password receive kar rahe hain
  const { name, email, password } = req.body;

  // Validation
  // Agar koi bhi field empty ho to error return karo
  if (!name || !email || !password) {

    return res.status(400).json({

      message: "All fields are required"

    });

  }

  // Duplicate Email Check
  // Agar same email pehle se array me mojud hai to registration allow nahi hogi
  const existingUser = users.find(user => user.email === email);

  if (existingUser) {

    return res.status(400).json({

```

```
message: "Email already exists" });
```

```
}
```

```
// Naya user object create kar rahe hain
```

```
const newUser = {
```

```
// Har user ko unique ID dene ke liye id:  
users.length + 1,
```

```
name,
```

```
email,
```

```
password
```

```
};
```

```
// User ko array me save kar rahe hain
```

```
users.push(newUser);
```

```
// Success response client ko bhej rahe hain
```

```
res.status(201).json({
```

```
message: "Registration Successful",
```

```
user: newUser
```

```
});
```

```
});
```

```
// =====
```

```
// Get All Users API
```

```
// =====
```

```
// GET request se tamam users ka data return
```

```
hoga app.get("/users", (req, res) => {
```

```
res.json(users);
```

```
});
```

```
// =====
```

```
// Start Server
```

```
// =====  
  
// Server ko given port par run kar rahe  
hain app.listen(PORT, () => {  
  
  console.log(`Server Running On Port ${PORT}`);  
});
```

Step 5 Run Server

```
node server.js
```

OUTPUT

```
Server Running On Port 3000
```

Step 6 Open Postman

POST

```
http://localhost:3000/register
```

BODY — JSON

```
{  
  "name": "Ali",  
  "email": "ali@gmail.com",  
  "password": "123456"  
}
```

EXPECTED OUTPUT

```
{  
  "message": "Registration Successful",  
  "user": {  
    "id": 1,  
    "name": "Ali",  
    "email": "ali@gmail.com",  
    "password": "123456"  
  }  
}
```

GET

```
http://localhost:3000/users
```

OUTPUT

```
[  
  {  
    "id": 1,  
    "name": "Ali",  
    "email": "ali@gmail.com",  
    "password": "123456"  
  }  
]
```

ASSIGNMENT

Create **Student Registration API**.

Fields

- ▶ Name
- ▶ Email
- ▶ Password

Endpoints

```
POST /register  
GET /users
```